

Original project description — verbatim

The text below preserves the original wording, spelling, numbering and examples. Diagrams follow on separate pages.

Our project has 3 components:

A. Database Table creation:

1. We will release a google form which will take the following inputs:

Name:

Student ID (if applicable)

Blood type: [drop down menu]

Department (if applicable): [drop down menu]

Last blood donated: [DD/MM/YY]

Geographic location: [drag & drop in google maps] (We will divide the entire map into small grid square (say each being 0.25km^2) & when the user inputs his location he will be placed in one of the grids)

2. We will make/take a dataset of all the hospitals of the country (or dhaka) & place them in our grid based map too

3. Now our backend software will take our user's grid & the grid of each hospital in the country & make an array of the distance between the centers of user's grid & hospital grids. Like [0.5 65 8 0.5] km

4. All of these information will be stored in a table somewhere (need clarification here (1))

B. Patient Input

1. We will have a facebook page where people will dump their blood requests. The format will be messy text like "B+ blood needed urgently for my relative in Apollo after her C-section. Please contact: 01xxxxxx (patient's husband)"

2. We will have a also make a table for this data. A human volunteer will manually extract data from this raw text. (It would've been better if we could've used AI to automate this process). He will collect the following data

Gender: [drop down: Male/Female]

Blood type: [drop down: A+/ O-,.....]

Hospital: [dd: list of all hospital]

Contact no:

2nd contact (if applicable):

Urgency: [dd: High/Mid/Low] (needs clarification here(2))

Litre needed: [dd: 1:10]

C. Web based application:

Each of our donor will download a web based application

The web based application will also have the feature to register new donors (Section A)

The UI would be simple

1st layer: Registered donor / new donor (new donor will take to our form in section A)

2nd layer (after clicking on registered donor): 1. Update Last donated 2. Update Address 3. Remove myself as a Donor 4. Temporarily unavailable

D. Integration:

Suppose in Section B our human volunteer has just completed a request.

1. Now our backend software will go to our donor database
2. Filter out donors by last donated more than 120 days ago & Temporarily unavailable
3. Take the co-ordinate of the grid of the patient's hospital & compare it with the array we created in Section A [0.5 65 8] for the filtered donors
4. Select the geographically 10 closest donors
5. Send to them the message "Blood requested at Apollo today, pls contact 019xxxxx"

He will have the option to select 1. Available 2. Not available

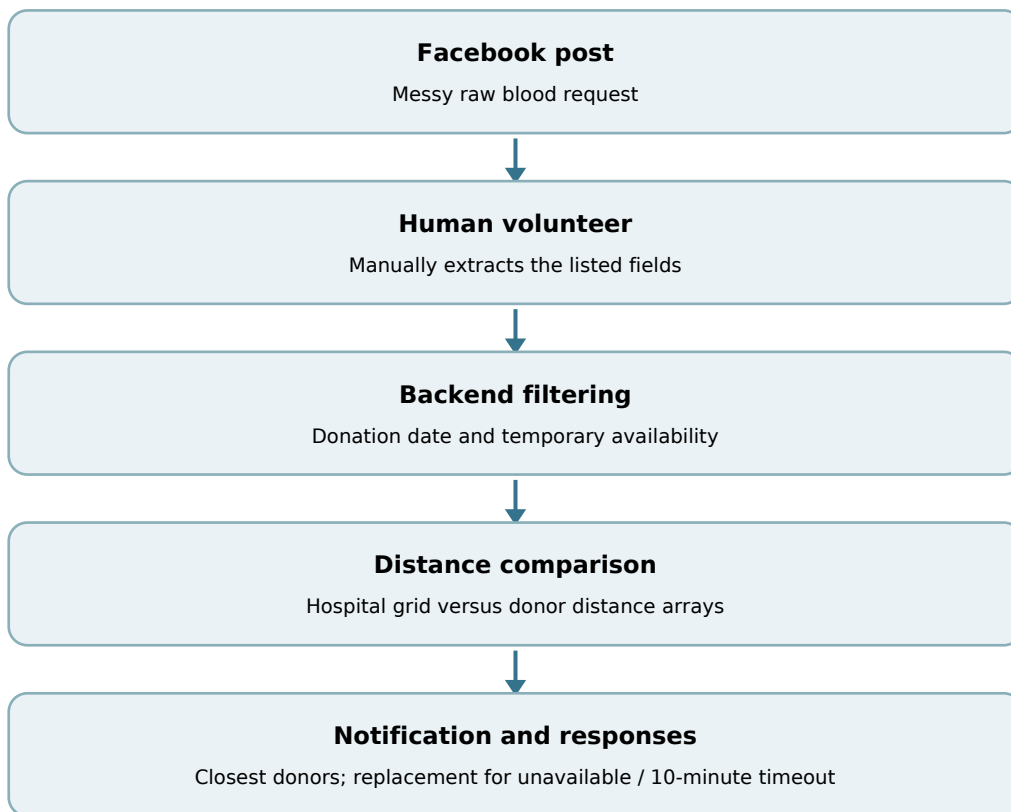
6. If anyone of them is not available the 11th closest person will send the message
7. Depending of the Litre needed (in Section B), usually 1L=1person, software will send message to appropriate number of donors.
8. For examples if 8 bags of blood required we will send message to first 8 donors. Suppose donor 1,2, 4,5 is unavailable, then we'll send message to 9,10,11, 12. This will continue until we reach 8 donors
9. If any donor will fail to respond within 10 minutes, he'll be considered unavailable

Additional diagrams (not part of the original wording)

A. Database creation

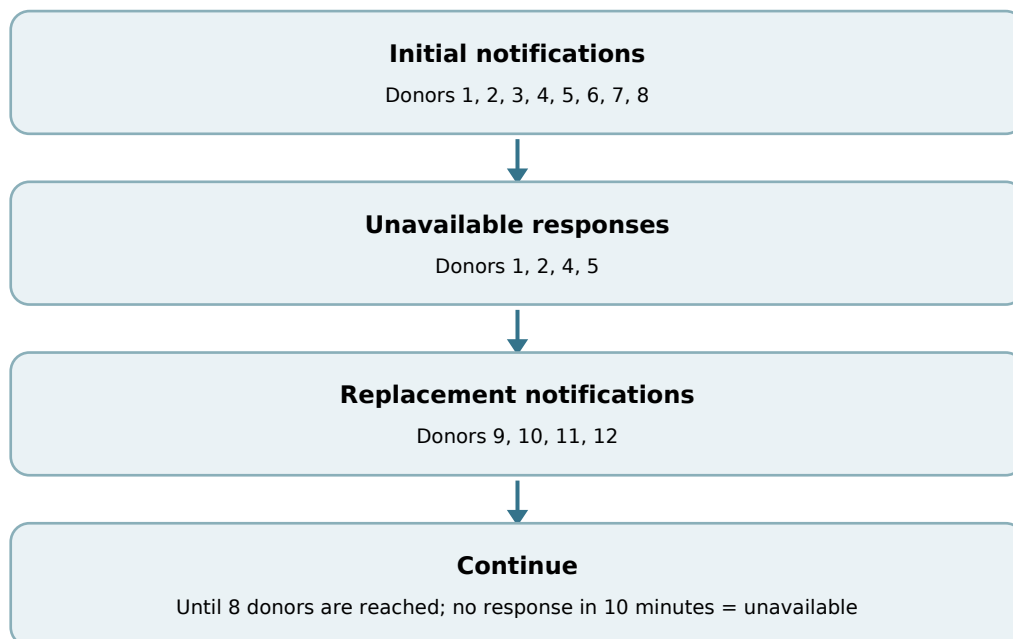


B → D. Request handling and matching



Example: 8 bags requested

Replacement sequence



C. Web based application

Donor interface

